

ORIGINAL ARTICLE

Analysis of CRISPR–Cas system and antimicrobial resistance in *Staphylococcus coagulans* isolates

A.C. Murugesan  and H.S. Varughese

Department of Veterinary Microbiology, Madras Veterinary College, Tamil Nadu Veterinary and Animal Sciences University, Chennai, India

Significance and impact of the study: *Significance:* The present study demonstrated the very high prevalence of the CRISPR–Cas system in *Staphylococcus coagulans* strains compared to other bacteria. The majority of the methicillin susceptible *S. coagulans* isolates contained Type IIC CRISPR–Cas system whereas many methicillin-resistant *S. coagulans* isolates had Type IIIA CRISPR–Cas system, and they were present within the SCCmec mobile genetic element. This distinct feature was not mentioned in other staphylococci. *Impact:* Detailed characterization of the CRISPR–Cas system in all coagulase positive and negative staphylococci will aid in the understanding of their importance in spreading/preventing mobile genetic elements among themselves and in antibiotic resistance transmission.

Keywords

antimicrobial resistant genes, CRISPR–Cas system, direct repeat sequences, spacers, *Staphylococcus coagulans*.

Correspondence

Ananda Chitra Murugesan, Department of Veterinary Microbiology, Madras Veterinary College, Tamil Nadu Veterinary and Animal Sciences University, Chennai 600007, India. E-mail: anandachitra.m@tanuvas.ac.in

2022/0051: received 30 January 2022, revised 5 March 2022 and accepted 30 March 2022

doi:10.1111/lam.13713

Abstract

CRISPR–Cas system contributes adaptive immunity to protect the bacterial and archaeal genome against invading mobile genetic elements. In this study, an attempt was made to characterize the CRISPR–Cas system in *Staphylococcus coagulans*, the second most prevalent coagulase positive staphylococci causing skin infections in dogs. Out of 45 *S. coagulans* isolates, 42/45 (93.33%) strains contained CRISPR–Cas system and 45 confirmed CRISPR system was identified in 42 *S. coagulans* isolates. The length of CRISPR loci ranged from 167 to 2477 bp, and the number of spacers in each CRISPR was varied from two spacers to as high as 37 numbers. Direct repeat (DR) sequences were between 30 and 37, but most (35/45) of the DRs contained 36 sequences. The predominant *S. coagulans* strains 29/45 did not possess any antimicrobial resistant genes (ARG); 26/29 strains contained Type IIC CRISPR–Cas system. Three isolates from Antarctica seals neither contain CRISPR–Cas system nor ARG. Only 15/45 *S. coagulans* strains (33.33%) harboured at least one ARG and 13/15 of them were having *mecA* gene. All the methicillin susceptible *S. coagulans* isolates contained Type IIC CRISPR–Cas system. In contrast, many (10/13) *S. coagulans* isolates which were methicillin resistant had Type IIIA CRISPR–Cas system, and this Type IIIA CRISPR–Cas system was present within the SCCmec mobile genetic element. Hence, this study suggests that Type II CRISPR–Cas in *S. coagulans* isolates might have played a possible role in preventing acquisition of plasmid/phage invasion and Type IIIA CRISPR–Cas system may have an insignificant role in the prevention of horizontal gene transfer of antimicrobial resistance genes in *S. coagulans* species.

Introduction

Bacteria acquire new traits that help them in metabolism, pathogenesis and antibiotic resistance through horizontal gene transfer (HGT) and thus, HGT is important in the evolution of the bacteria. However, bacteria are under

constant threat that the nonspecific integration of mobile genetic elements may result in lethal mutation or other deleterious effects on bacteria. Therefore, bacteria have also been developing a plethora of strategies that interfere with the HGT. CRISPR (clustered regularly interspaced short palindromic repeats)—Cas (CRISPR associated

genes) system is one such mechanism that acts as a protective barrier in prokaryotes that function in the adaptive immunity to mobile genetic elements. CRISPR–Cas discriminate self-DNA from non-self-DNA and destroy/cleave the non-self plasmid/virus. This is found in approximately 50% of a bacterial genome and 87% of archaeal genomes (Makarova *et al.* 2015).

Studies on the significance of the CRISPR–Cas system on antibiotic resistance and virulence exhibited dubious correlation in various bacteria. The presence of CRISPR–Cas system in *Enterococcus* (Tremblay *et al.* 2013), *Klebsiella pneumoniae* (Mackow *et al.* 2019), *Staphylococcus epidermidis* (Marraffini and Sontheimer 2008) and *Escherichia coli* K12 (Touchon *et al.* 2011) strains were less resistant to many antimicrobials. On the other hand, the CRISPR type II system present in *Campylobacter jejuni* plays an important role in the attachment and invasion of these bacteria in the colorectal epithelial cell model (Shabbir *et al.* 2016) and play potential involvement in enhancing the virulence of *C. jejuni* (Shabbir *et al.* 2018).

Staphylococcus coagulans is earlier identified as *Staphylococcus schleiferi* subspecies *coagulans* (Igimi *et al.* 1990) and recently, it has been recognized as a new species *S. coagulans* (Madhaiyan *et al.* 2020). *Staphylococcus coagulans* is the second most common cause of skin infections in dogs. In this study, we examined 45 *S. coagulans* whole-genome sequences for the presence of CRISPR–Cas system and characterized the direct repeats (DRs), spacers and Cas proteins. Further, attempts were made to understand the correlation between the presence of CRISPR, their types and antimicrobial resistance genes in *S. coagulans* isolates.

Results and Discussion

CRISPR locus in *S. coagulans* isolates

In the total 42 *S. coagulans* isolates in the NCBI GenBank and three *S. coagulans* clinical isolates whole-genome sequences, 42/45 (93.33%) strains contained CRISPR–Cas system and three isolates from healthy Antarctic seals collected in late 20th century did not contain CRISPR–Cas system. There were 45 confirmed CRISPR system identified in 42 *S. coagulans* isolates. *S. coagulans* isolates and strain-wise confirmed CRISPR loci related details are given in Table 1.

The prevalence of 93.33% CRISPR–Cas system was observed in *S. coagulans* isolates in this study. This was extremely very high compared to the analysis of the complete CRISPR–CAS systems in the CRISPR database of *S. aureus* with 0.08% (Zhao *et al.* 2018) and 4% in GenBank submitted *S. epidermidis* whole-genome sequences (Li *et al.*

2016). However, 14% of the clinical isolates of *S. epidermidis* possessed CRISPR–Cas system (Li *et al.* 2016). A relatively higher prevalence of 37% of *K. pneumoniae* strains carried Type IE or IE*CRISPR–Cas on their chromosome, and 5% of the *K. pneumoniae* plasmid contained Type IV like CRISPR–Cas system (Kamruzzaman and Iredell 2020).

Among the 45 confirmed CRISPR loci, 40 *S. coagulans* isolates comprised only one confirmed CRISPR locus, and three (strain TSCC54) and two CRISPR loci (strain 196) were recognized in each one isolates (Table 1). Our study corroborates the recent research in *S. aureus* in which most of the isolates contained only one CRISPR locus, and few strains only had multiple loci (Zhao *et al.* 2018).

Length of CRISPR loci ranged from 167 to 2477 bp, and the number of spacers in each CRISPR was varied from two spacers to as high as 37 (Table 1). DR sequences were between 30 and 37, but most (35/45) of the DRs contained 36 sequences (Table S2).

DR sequences

The conservation percentage of DR sequences within the CRISPR of *S. coagulans* isolates was at least 80% which means that the DR sequences in the same CRISPR system are similar or identical. Based on the consensus DR sequences, 45 *S. coagulans* CRISPR loci were grouped into 10 categories (Table 2), and Group 2 was the predominantly ($n = 21$) seen consensus DR. Group 5 DR sequence was observed in 12 *S. coagulans* isolates. In contrast, five consensus DR sequences were seen in only one of the *S. coagulans* isolate. The strain-wise DR sequences details are given in Table S2.

Though DR sequences in the *S. coagulans* isolates were highly conserved, they were not 100% identical throughout CRISPR locus. Generally, repeat sequences within loci are not highly conserved, and frequent loss of last terminal repeat was seen (Horvath *et al.* 2008). This was observed in group 6 sequence, which was the part of group 1 and group 8 DR sequences but last 6/7 bases were lost. Group 2 and group 7 DR sequences were similar except that “A” was added as first base in the group 7.

Typically, the number of repeats is an indicator of the functionality and integrity of CRISPR–Cas system. If the number of repeats is high, generally, CRISPR–Cas system is complete. When the number of repeats is intermediate, the erosion of the Cas system is recent, and when the repeats are few, only relics of the system are noticed (Touchon and Rocha 2010). In this study, 19 *S. coagulans* isolates contained 5 or less than 5 DR sequences.

RNA secondary structure of DR sequences of these 10 groups predicted by RNAfold Webserver revealed the

Table 1 Details of CRISPR loci in *Staphylococcus coagulans* isolates

S. no.	Strain	Host	GenBank ID	No. of confirmed CRISPR	No. of spacers	CRISPR length	DR length	CRISPR start & end position/ Contig No.
1	TSCC54*	Dog	AP014944	3	16,15,3	1178; 1023;251	36; 36;37	66853-68031; 1549038-1550061; 76758-77009
2	2317-03	Dog	CP010309	1	19	1290	36	1585815-1587105
3	OT1-1	Dog	CP035007	1	17	1160	36	15668-16828
4	5909-02	Dog	CP009676	1	16	1091	36	1544989-1546080
5	1360-13	Dog	CP009470	1	37	2477	36	1478814-1481291
6	2142-05	Dog	CP009762	1	5	365	36	1478868-1479233
7	196*	Human	POVH01	2	3,6	252;429	36;36	44645-44897: contig-2; 61687-62116: contig-6
8	DSM 6628	Dog	PPQN01	1	18	1221	36	1460-2681
9	66170	Dog	JAABPE01	1	15	1026	36	550084-551110; contig-1
10	65195	Dog	JAABPF01	1	4	299	36	507303-507602; contig-1
11	M31/11/1	Grey Seal	JABTCO01	1	2	167	36	167342-167509; contig-1
12	M615/02/4	Grey Seal	JABTCN01	1	3	232	36	1872-2104; contig-57
13	M611545/16/1	Grey Seal	JABTCP01	1	14	957	36	5632-6589; contig-62
14	1028910	Dog	VFBN01	1	17	1156	36	254601-255757; contig-80
15	1031373	Dog	Yet to be assigned	1	6	456	30	191175-191630; contig-3
16	1031336	Dog	Yet to be assigned	1	9	632	37	47466-48097; contig-10
17	A/G16/00/1	Seal	JAABPB01	0	Nil	Nil	Nil	Nil
18	A/G14/99/8	Seal	JAABPD01	0	Nil	Nil	Nil	Nil
19	A/W41/99/1	Seal	JAABPC01	0	Nil	Nil	Nil	Nil
20	p1691527012	Dog	JABBLT01	1	4	300	36	436434-436733; Node 2
21	94832	Dog	JABBL01	1	4	300	36	436375-436674; Node 1
22	p1788028585	Dog	JABBLV01	1	8	564	36	436054-436617; Node 1
23	88276	Dog	JABBLF01	1	7	498	36	804873-805370; Node 1
24	94233	Dog	JABBLK01	1	4	300	36	759689-759988; Node 1
25	79461	Dog	JABBLB01	1	8	564	36	786459-787022; Node 1
26	83390	Dog	JABBLD01	1	2	168	36	759689-759856; Node 1
27	79099	Dog	JABBKZ01	1	6	432	36	436434-436865; Node 1
28	79320	Dog	JABBLA01	1	7	535	37	122282-122816; Node 5
29	p64509910	Dog	JABBLS01	1	5	386	30	296-681; Node 14
30	p30834477	Dog	JABBLQ01	1	5	386	30	122274-122659; Node 5
31	78868	Dog	JABBK01	1	5	391	35	37007-37397; Node 10
32	89507	Dog	JABBLH01	1	3	248	33	27964-28211; Node 10
33	p53098117	Dog	JABBLR01	1	5	391	35	37007-37397; Node 5
34	p1135317838	Dog	JABBLN01	1	8	564	36	759687-760250; Node 1
35	89657	Dog	JABBLI01	1	2	168	36	758749-758916; Node 1
36	p15691995	Dog	JABBL001	1	8	564	36	10945-11508; Node 1
37	78830	Dog	JABBLP01	1	4	300	36	436307-436606; Node 1
38	p1738727807	Dog	JABBLU01	1	4	300	36	436307-436606; Node 1
39	88282	Dog	JABBLG01	1	4	300	36	772879-773178; Node 1
40	81701	Dog	JABBLC01	1	4	300	36	436469-436768; Node 1
41	91171	Dog	JABBLJ01	1	7	535	37	3427-3961; Node 12
42	M2043/98/1	Dog	JAGFWY01	1	14	960	36	45518-46477; Node 4
43	86039	Dog	JABBLE01	1	9	630	36	45518-46147; Node 2
44	91460c	Dog	JABBLM01	1	4	300	36	9090-9389; Node 2
45	p17922383	Dog	JABBLP01	1	13	894	36	45447-46340; Node 15

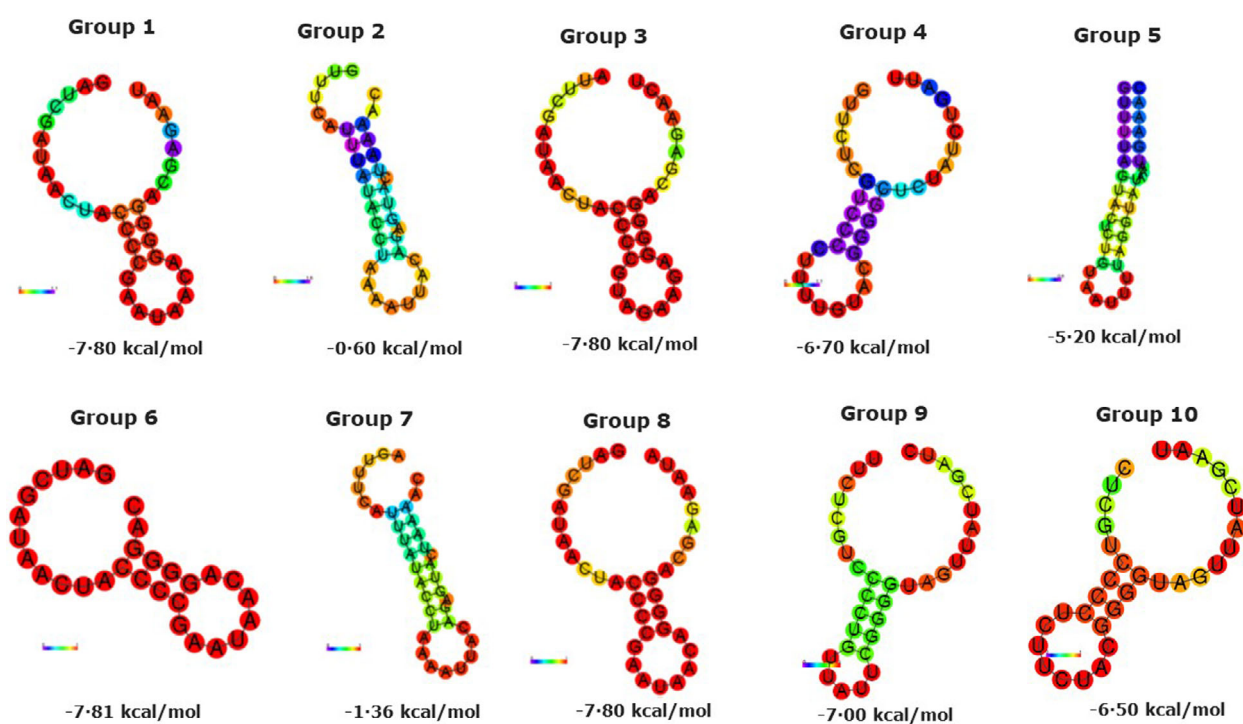
The presence of more than one number of confirmed CRISPR-Cas system in the bacterial strain was indicated in bold.

Abbreviation: DR, direct repeat.

*Strain contains more than one CRISPR-Cas system.

Table 2 Direct repeat sequence and RNA secondary structure minimum free energy (MFE) value of CRISPR of *Staphylococcus coagulans* isolates

Group	Direct repeat consensus sequences	Length	GC %	No. of Strain(s)	MFE value (kcal/mol)
1	GATCGATAACTACCCGAATAACAGGGGACGAGAAT	36	34	1	-7.80
2	GTTTCATTATACCTAAAATTACAGAGTACTAAAC	36	25	21	-0.60
3	ATTCGATAACTACCCGTAAGAAGAGGGGACGAGAACT	37	49	1	-7.80
4	GTTCTCGTCCCTTTGTACGGGGCTCTATCTGATT	36	50	1	-6.70
5	GTTTTAGTACTCTGTAATTTAGGTATAAATGAAAC	36	25	12	-5.20
6	GATCGATAACTACCCGAATAACAGGGGAC	30	50	3	-7.81
7	AGTTTCATTATACCTAAAATTACAGAGTACTAAAC	37	24	1	-1.36
8	GATCGATAACTACCCGAATAACAGGGGACGAGAATA	37	46	2	-7.80
9	TTCTCGTCCCTGTTATTCGGGGTAGTTATCGATC	35	49	2	-7.00
10	CTCGTCCCTCTTCTACGGGGTAGTTATCGAAT	33	52	1	-6.50

**FIGURE 1** RNA secondary structure prediction and minimum free energy (MFE) values of Direct Repeat Groups of *Staphylococcus coagulans* isolates

typical loop-stem structure of RNA in all the groups (Fig. 1). Generally, two loops connected by a stem structure were observed, except for group 5, which had only one loop but with a very long stem (13 bp). A stem length of 4 bp was noted in groups 1, 3, 6, 8 and 10, whereas 9 bp stem length was noticed in groups 2 and 7. A stem length of 5 was seen in groups 4 and 9.

Repeat sequences form a part of crRNA (CRISPR RNA), and the normal activity of RNA is determined by its secondary structure. Minimum free energy (MFE) algorithm is one of the computer-based tools to predict

the secondary structure of RNA. The number of the sequences, GC content and their arrangement generally affect the MFE of an RNA molecule (Zhao *et al.* 2018). If the sequences are long, they form more stacking and hydrogen bond interaction, and GC rich RNAs are more stable than AU-rich sequences. Nucleotide order influences the folding structure stability, because it determines the number and the extension of loops and double-helix conformations (Trotta 2014). The present study also noticed that high GC content repeat sequences groups (3, 4, 6, 8, 9 and 10) had low MFE values.

MFE value was found to be highest for group 2 and lowest to be for group 6. A low MFE value was also noticed for groups 1, 3, 4, 8, 9 and 10, and an intermediate value was noted for group 5, whereas groups 2 and 7 had high MFE values, which means that they were highly unstable compared with other groups.

Spacers

In total, there were 378 spacers in 45 confirmed CRISPR in 42 *S. coagulans* strains. Among 378 spacers, 128 spacers were unique, and 70 spacers were shared with at least one other strain resulted in a sum of 198 spacers in all *S. coagulans* isolates (Fig. 2). Details of the strain-wise spacer sequences are given in Table S3. All the spacers in *S. coagulans* strain 196, M611545/16/1, M31/11/1 and 89507 were completely unique, whereas all the 17 spacers present in *S. coagulans* strain OT1 were shared with other strains. Most frequently seen spacers were GGGCTAC TAATCTTTTGAAGTCTGCATAGT and ACCAGCCACC

AGGTTGGAACAGATACCCTA, which was found in 13 and 11 various strains respectively. The spacers TTA-TAATTTCAATGTCGTGCGCAATCAATT and TCCTACC TCAAGTGCTACGGCTTTAACAAA were present in 10 different strains each. Among these four spacers sequences, last three spacers' sequences had homologous with phage sequences.

The maximum number of 37 spacers was seen in a single CRISPR locus in strain 1360-13, whereas only two spacers were observed in strains M31/11/1, 83390 and 89657. The CRISPR locus of six different patterns, in terms of CRISPR locus length, same spacers and DR sequences were seen in among some strains of the *S. coagulans*. They were p1691527012=94832=78830=p173872780=81701, 94233 = 88282=91460c, 79461=p113531783=p15691995, 79320 = 91171, p64509910=p30834477 and 78868=p5309811. These identical patterns are shown with same spacer numbers/colour/patterns in Fig. 2, and with the same colours in Fig. S1.

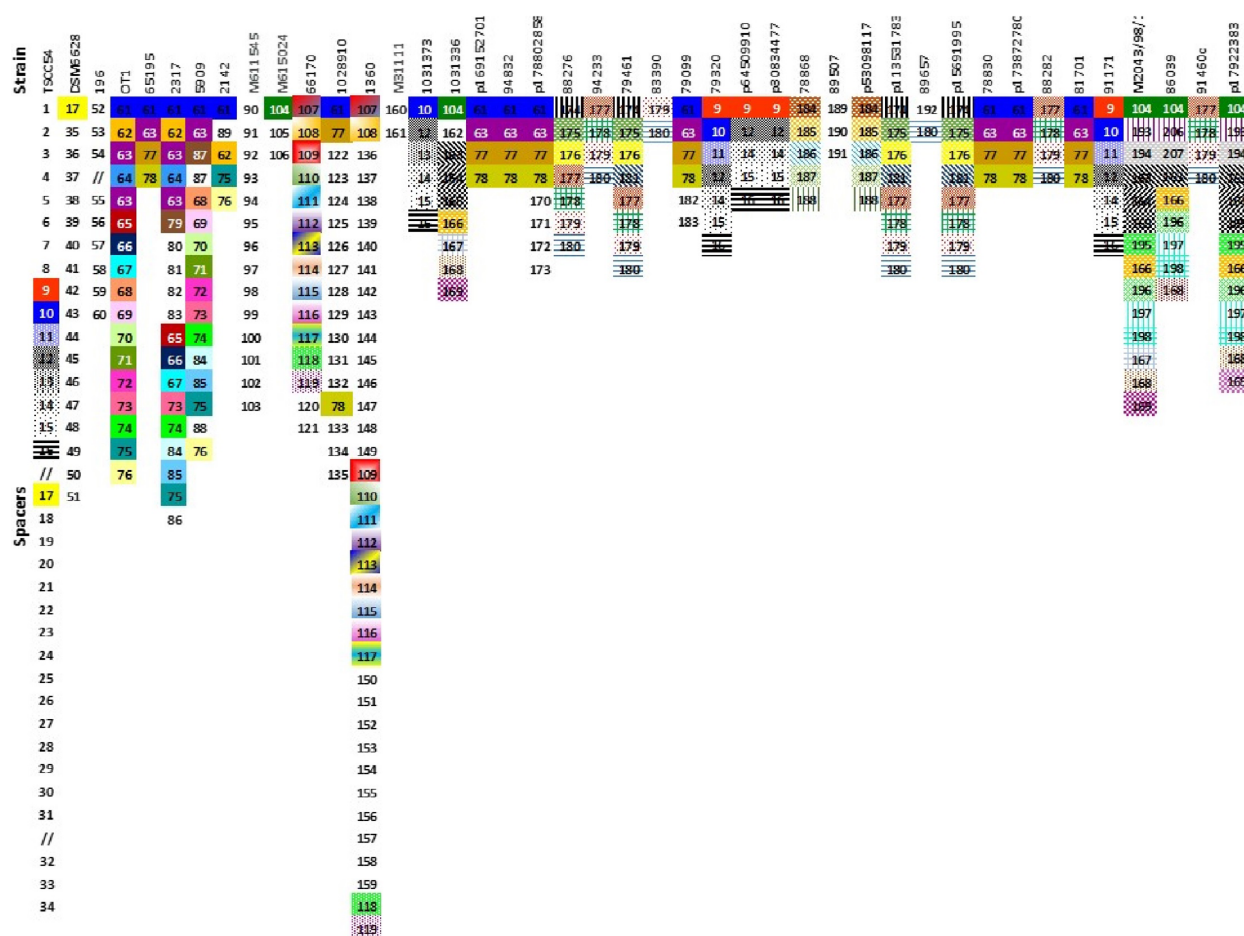


FIGURE 2 A spacer map for the distribution of spacers in CRISPR loci of *Staphylococcus coagulans* isolates. Unique spacers are represented in box without colour or pattern. Identical spacers are represented by the same number and colour or pattern

Strains p1788028585 and 79099 had same spacers with one of the identical CRISPR but both the strains acquired new and different spacers near their leader sequences. It is worth mentioning that identical CRISPR loci were seen within methicillin resistance and within methicillin susceptible *S. coagulans* strains but not between methicillin resistance and susceptible *S. coagulans* strains. Moreover, identical CRISPR noticed in *S. coagulans* strains were isolated from the same country and host, but date of collection varies.

Further analysis of NCBI Blast search of all spacer sequences revealed that altogether 27 and 67 spacer sequences matched the corresponding sequences in plasmid and phages respectively (Tables S4 and S5). Most of these plasmids were from staphylococci, but other plasmids ranged from Gram-positive enterococci, *Rhodococcus*, clostridial species to Gram-negative *Klebsiella*, *Enterobacter*, *Borrelia*, *Burkholderia* species. Similarly, the majority of the phages' sequences were from staphylococci. However, phage sequences of *Streptococcus*, *Mycobacteria*, *Vibrio*, *Bacillus*, *Klebsiella*, *Enterobacter*, *Lactobacillus*, *Escherichia* and *Pelagibacter* were also noticed. Six spacer sequences were matched with both plasmid and phage sequences, and some of these sequences were matched with UTP pyrophosphatase and phage major capsid protein.

It is trustworthy to mention that some spacers were 100% identical with *S. coagulans* chromosome sequences but not their current host bacteria. There were no matching sequences for more than 50% (108/198) of the spacers in the GenBank database.

CRISPR types and *cas* genes in *S. coagulans* CRISPR system

Staphylococcus coagulans strains contained Class I CRISPR Type IIIA and Class II Type IIC systems (given with same colour pattern in Fig. S1). *Staphylococcus coagulans* strains TSCC54 and 196 contained both Type IIIA and Type IIC *cas* genes. On the other hand, all other strains exclusively contained either Type IIC or Type IIIA. Majority of the methicillin-resistant *S. coagulans* strains contained Type IIIA, whereas methicillin susceptible strains contained Type IIC CRISPR–Cas system. It has been noticed that all *cas* genes were located downstream to CRISPR array in both Type II and III systems in *S. coagulans* isolates. The structural arrangements of *cas* genes with/without other associated proteins are depicted in the Fig. S2.

The *cas1* and *cas2* genes, which are vital components of the active CRISPR system, were found in 42 out of 45 confirmed CRISPR locus in 42 *S. coagulans* strains. Among the three CRISPR loci present in *S. coagulans* strain TSCC54, only two CRISPR loci contained *cas1* and *cas2* genes. It is interesting to note that *cas1* gene in Type IIIA CRISPR loci in strain 196 and Type IIC in strain

M611545/16/1 were frameshifted resulted in inactive CRISPR–Cas system in these strains. A frameshift mutation was also noticed in the *cas9* gene of strains M2043/98/1, 86039 and p17922383.

It was found that five *S. pyogenes* strains contained CRISPR–Cas system without *cas* genes, and they had more numbers of prophages than other *S. pyogenes* strains. Spacer analysis revealed that one spacer targeted a protospacer on its own chromosome in four out of five *S. pyogenes* isolates. This demonstrated that the limited presence and activity of CRISPR–Cas systems in *S. pyogenes* had allowed the introduction of virulence genes by phages into *S. pyogenes*, thereby contributing to the strain-specific pathogenicity, that is the characteristic of this species (Nozawa *et al.* 2011).

CRISPR–Cas system without active *cas1* gene was seen in *S. coagulans* strains 196 and M611545/16/1. Since the *cas1* gene encodes a nuclease, which is involved in spacer acquisition (Yosef *et al.* 2012), a frameshift mutation in the *cas1* gene in these strains may indicate that they lost their potential for CRISPR adaptation as demonstrated in *Enterococcus faecium* (van Schaik *et al.* 2010). On the other hand, it cannot be ruled out that CRISPR interference may still be effective despite the loss of function of the *cas1* gene (Louwen *et al.* 2014). It should be noted that strain 196 still contained a fully active Type IIC CRISPR–Cas system, and hence, it may still acquire new spacers through it.

It has been noted that increased antibiotic resistance and uptake of phages and pathogenicity island in *E. faecium* were associated with deletion of *cas* genes (van Schaik *et al.* 2010). Similarly, a frameshift mutation in the *cas1* gene in strains 196 and M611545/16/1 may pave the way for this strain to acquire new resistance genes through plasmid or phages or other mobile genetic elements, such as SCCmec.

Identical CRISPR arrays in some *S. coagulans* strains implying that there might have been a HGT between these strains. It is interesting that all the six identical CRISPR pattern containing strains were having active CRISPR–Cas system. Some *S. coagulans* strains contained shared spacers with identical CRISPR pattern but had acquired additional spacer sequences. Hence, though the horizontal transfer of CRISPR occurs within or among staphylococci species, the exact copy is not transmitted, making them a complicated and diversified system. This can be tapped for typing of staphylococci isolates in the future.

CRISPR–Cas system, SCCmec and antibiotic resistance genes

The predominant *S. coagulans* strains 29/45 did not possess any antimicrobial resistant genes (ARG); 26/29

isolates contained Type IIC CRISPR–Cas system. Three isolates collected in late 20th Century from Antarctica seals neither contain CRISPR–Cas system nor ARG. Only 15/45 *S. coagulans* strains (33.33%) harboured at least one ARG and 13/15 of them were having *mecA* gene. Details of the *S. coagulans* isolates and ARG in their genome are presented in the Fig. S1. Majority of the *S. coagulans* (10/15) strains with ARG were harbouring Type IIIA CRISPR–Cas system. Recently, Mo *et al.* (2021) demonstrated that the nonspecific DNase activity of the staphylococcal Type IIIA CRISPR–Cas system increases mutations in the host and accelerates the generation of antibiotic resistance in *S. aureus* and *S. epidermidis*.

Majority of the *S. coagulans* isolates which contained Type IIC system did not contain any ARG. Only two isolates (p15691995 and M2043/98/1) with Type IIC CRISPR–Cas system were possessing only one ARG (β -lactamase) in their genome, and strain p15691995 contained unusual phenotype bla_{TEM-1 β} . This suggests that Type IIC CRISPR–Cas system in *S. coagulans* strains may play a possible role in inhibiting the acquisition of ARG containing mobile genetic elements. However, this necessitates additional studies to prove the possible role of Type IIC CRISPR–Cas in the virulence and antibiotic resistance in this species.

In the present study, 13 *S. coagulans* strains harboured SCCmec mobile genetic element, which had methicillin resistance genes along with resistance to few heavy metals. It was observed that with the exception of OT1, 2317 and 5909 strains, all other strains possessed a Type IIIA CRISPR–Cas system within their SCCmec region. Similar to the presence of CRISPR–Cas system in SCCmec region in *S. coagulans* strains, it was also identified in *S. aureus* strains (Kinnevey *et al.* 2013) and in *Staphylococcus pseudintermedius* strains (McCarthy *et al.* 2015; Duim *et al.* 2018). The present study observation along with the studies of McCarthy *et al.* (2015) and Mo *et al.* (2021) suggest that Type IIIA CRISPR–Cas system might have an insignificant role in the prevention of HGT in staphylococcal species.

A recent study of CRISPR–Cas analysis in different staphylococci species and spacer sequence analysis within the closely related types demonstrated that *S. coagulans* strain TSCC54 and *S. aureus* strain 80BA02176 type IIIA in the SCCmec region might have been acquired from *S. capitis* strain CR01 (Rossi *et al.* 2017). It has been thought that the SCCmec and type IIIA CRISPR–Cas system were originally acquired from coagulase negative staphylococci which act as a pool of genetic resources for coagulase positive staphylococci such as *S. aureus*, *S. pseudintermedius* and *S. coagulans* (Rossi *et al.* 2017).

Eight *S. coagulans* strains contained additional ARG other than *mecA*, and interestingly, these ARG were

located on transposable elements and generally clustered together as predicted by pathogenicity island viewer. *S. coagulans* strain 1031373 was the only multidrug resistance strain by possessing ARG against five different classes of antibiotics (shown in Fig. S1), which were located on two large transposable elements (Fig. S3). McCarthy *et al.* (2015) revealed that many resistant genes apart from SCCmec in multidrug methicillin-resistant *S. pseudintermedius* isolates were localized on transposable elements, and all genomes carried at least two transposons or insertion sequence elements.

The majority of the methicillin susceptible *S. coagulans* isolates contained Type IIC CRISPR–Cas system and they did not possess any antimicrobial resistance genes. This suggests that Type IIC CRISPR–Cas system in *S. coagulans* strains may inhibit the acquisition of ARG containing mobile genetic elements. In contrast, many *S. coagulans* isolates which were methicillin resistant had Type IIIA CRISPR–Cas system, and this Type IIIA CRISPR–Cas system was present within the SCCmec mobile genetic element. The present study observation suggests that Type IIIA CRISPR–Cas system may play an insignificant role in the prevention of HGT of antimicrobial resistance genes in *S. coagulans* species.

Materials and methods

Staphylococcus coagulans clinical isolates

Staphylococcus coagulans strains 1028910, 1031323 and 1031336 were isolated from skin infections of dogs from the previous study conducted during 2017–2018 (data not shown), and were maintained in the department. Genomic DNA of these clinical isolates were extracted using Xcelgen bacterial gDNA isolation kit and quality was assessed by agarose gel electrophoresis and Qubit 2.0 Fluorometer was used to determine the DNA concentration. Whole-genome sequencing was carried out using Illumina© technology in a commercial firm (Xcelris Labs Limited, Gujarat, India) to generate 2 × 150 bp paired end FASTQ data. Genome assembly was performed using Unicycler version 0.4.8.0 (Wick *et al.* 2017). ABACAS version 1.3.1 was used to arrange the contigs based on *S. coagulans* TSCC54 as reference strain (Assefa *et al.* 2009). Genome annotation was carried out using Prokka 1.14.5, NCBI Prokaryotic Genome Annotation Pipeline (Tatusova *et al.* 2016), and Rapid Annotations using Subsystems Technology (RAST) annotation pipeline (Aziz *et al.* 2008). The assembled genomes were submitted to GenBank, NCBI database, and accession No. for 1028910 is VFBN01. Accession numbers for the other two genomes are yet to be assigned (*S. coagulans* strain 1031323 BioProject PRJNA734103; Biosample SAMN19471632 and

strain 1031336 BioProject PRJNA734121, Biosample SAMN19471816).

Data sources

All the 42 *S. coagulans* isolates genomes submitted in the GenBank, NCBI, USA (last browsed on December 31, 2021) along with three *S. coagulans* clinical isolates' whole-genome sequences had been given as input to search for CRISPR loci and *cas* genes in CRISPR Finder server (<http://crispr.i2bc.paris-saclay.fr/Server/>) (last updated on January 10th 2022). Details of the *S. coagulans* isolates used in this study are given in Table S1.

Analysis methods

DR sequences of all confirmed CRISPR loci were grouped together based on the similarity sequences. RNA secondary structures and MFE of each DR sequence were predicted by the RNAfold Webserver (<http://rna.tbi.univie.ac.at/cgi-bin/RNAWebSuite/RNAfold.cgi>) by keeping all options in default settings (Mathews *et al.* 2004). Spacer sequences homology were searched in BLASTN of NCBI, USA, and the program selection was optimized for somewhat similar sequences. Furthermore, *cas* genes in the proximity of the CRISPR were explored in CRISPRs database (<http://crispr.i2bc.paris-saclay.fr/crispr/>). Antimicrobial resistance profile of the *S. coagulans* isolates was performed using ResFinder tool in the Centre for Genomic Epidemiology website (<https://cge.cbs.dtu.dk/services/ResFinder/>). Genomic island was predicted in IslandViewer 4 (<https://www.pathogenomics.sfu.ca/islandviewer/>).

Acknowledgement

The authors are thankful to Tamil Nadu Veterinary and Animal Sciences University Chennai, India for providing necessary infrastructure to carry out the research work.

Conflict of Interest

The authors declare no conflict of interest.

Funding

Funds for this study were received under DST-SERB project, Government of India (EMR/2016/006141).

Data Availability Statement

All relevant data are within the manuscript and in the supplementary tables and figure files.

References

- Assefa, S., Keane, T.M., Otto, T.D., Newbold, C. and Berriman, M. (2009) ABACAS: algorithm-based automatic contiguation of assembled sequences. *Bioinformatics* **25**, 1968–1969.
- Aziz, R.K., Bartels, D., Best, A.A., DeJongh, M., Disz, T., Edwards, R.A., Formsma, K., Gerdes, S. *et al.* (2008) The RAST Server: rapid annotations using subsystems technology. *BMC Genom* **9**, 75.
- Duim, B., Verstappen, K., Kalupahana, R.S., Ranathunga, L., Fluit, A.C. and Wagenaar, J.A. (2018) Methicillin-resistant *Staphylococcus pseudintermedius* among dogs in the description of novel SCCmec variants. *Vet Microbiol* **213**, 136–141.
- Horvath, P., Romero, D.A., Coûté-Monvoisin, A.C., Richards, M., Deveau, H., Moineau, S. *et al.* (2008) Diversity, activity, and evolution of CRISPR loci in *Streptococcus thermophilus*. *J Bacteriol* **190**, 1401–1412.
- Igimi, S., Takahashi, E. and Mitsuoka, T. (1990) *Staphylococcus schleiferi* subsp. *coagulans* subsp. nov., isolated from the external auditory meatus of dogs with external ear otitis. *Int J Syst Bacteriol* **40**, 409–411.
- Kamruzzaman, M. and Iredell, J.R. (2020) CRISPR-Cas system in antibiotic resistance plasmids in *Klebsiella pneumoniae*. *Front Microbiol* **10**, 2934.
- Kinnevey, P.M., Shore, A.C., Brennan, G.I., Sullivan, D.J., Ehrlich, R. *et al.* (2013) Emergence of sequence type 779 methicillin-resistant *Staphylococcus aureus* harboring a novel pseudostaphylococcal cassette chromosome mec (SCCmec)-SCC-SCCCRISPR composite element in Irish hospitals. *Antimicrob Agents Chemother* **57**, 524–531.
- Li, Q., Xie, X., Yin, K., Tang, Y., Zhou, X., Chen, Y., Xia, J., Hu, Y. *et al.* (2016) Characterization of CRISPR-Cas system in clinical *Staphylococcus epidermidis* strains revealed its potential association with bacterial infection sites. *Microbiol Res* **193**, 103–110.
- Louwen, R., Staals, R.J.H., Endtz, H.P., van Baarlen, P. and van der Oost, J. (2014) The Role of CRISPR-Cas Systems in Virulence of Pathogenic Bacteria. *Microbiol Mol Biol Rev* **78**, 74–88.
- Mackow, N.A., Shen, J., Adnan, M., Khan, A.S., Fries, B.C. and Diago-Navarro, E. (2019) CRISPR-Cas influences the acquisition of antibiotic resistance in *Klebsiella pneumoniae*. *PLoS One* **14**, e0225131.
- Madhaiyan, M., Wirth, J.S. and Saravanan, V.S. (2020) Phylogenomic analyses of the *Staphylococcaceae* family suggest the reclassification of five species within the genus *Staphylococcus* as heterotypic synonyms, the promotion of five subspecies to novel species, the taxonomic reassignment of five *Staphylococcus* species to *Mammaliococcus* gen. nov., and the formal assignment of *Nosocomiicoccus* to the family *Staphylococcaceae*. *Int J Syst Evol Microbiol* **70**, 5926–5936.

- Makarova, K.S., Wolf, Y.I., Alkhnbashi, O.S., Costa, F., Shah, S.A., Saunders, S.J., Barrangou, R., Brouns, S.J.J. *et al.* (2015) An updated evolutionary classification of CRISPR–Cas systems. *Nat Rev Microbiol* **13**, 722.
- Marraffini, L.A. and Sontheimer, E.J. (2008) CRISPR interference limits horizontal gene transfer in staphylococci by targeting DNA. *Science* **322**, 1843–1845.
- Mathews, D.H., Disney, M.D., Childs, J.L., Schroeder, S.J., Zuker, M. and Turner, D.H. (2004) Incorporating chemical modification constraints into a dynamic programming algorithm for prediction of RNA secondary structure. *Proc Nat Acad Sci USA* **101**, 7287–7292.
- McCarthy, A.J., Harrison, E.M., Stanczak-Mrozek, K., Leggett, B., Waller, A. *et al.* (2015) Genomic insights into the rapid emergence and evolution of MDR in *Staphylococcus pseudintermedius*. *J Antimicrob Chemother* **70**, 997–1007.
- Mo, C.Y., Mathai, J., Rostøl, J.T., Varble, A., Banh, D.V. and Marraffini, L.A. (2021) Type III-A CRISPR immunity promotes mutagenesis of staphylococci. *Nature* **592**, 611–615.
- Nozawa, T., Furukawa, N., Aikawa, C., Watanabe, T., Haobam, B., Kurokawa, K., Maruyama, F. and Nakagawa, I. (2011) CRISPR inhibition of prophage acquisition in *Streptococcus pyogenes*. *PLoS One* **6**, e19543.
- Rossi, C.C., Souza-Silva, T., Araújo-Alves, A.V. and Giambiagi-deMarval, M. (2017) CRISPR-Cas systems features and the gene-reservoir role of coagulase-negative Staphylococci. *Front Microbiol* **8**, 1545.
- Shabbir, M.A.B., Hao, H., Shabbir, M.Z., Hussain, H.I., Iqbal, Z., Ahmed, S., Sattar, A., Iqbal, M. *et al.* (2016) Survival and evolution of CRISPR–Cas system in prokaryotes and its applications. *Front Immunol* **7**, 375.
- Shabbir, M.A.B., Tang, Y., Xu, Z. *et al.* (2018) The involvement of the *Cas9* gene in virulence of *Campylobacter jejuni*. *Front Cell Infect Microbiol* **8**, 1–12.
- Tatusova, T., DiCuccio, M., Badretdin, A., Chetvernin, V., Nawrocki, E.P., Zaslavsky, L., Lomsadze, A., Pruitt, K.D. *et al.* (2016) NCBI prokaryotic genome annotation pipeline. *Nucleic Acids Res* **44**, 6614–6624.
- Touchon, M., Charpentier, S., Clermont, O., Rocha, E.P., Denamur, E. and Branger, C. (2011) CRISPR distribution within the *Escherichia coli* species is not suggestive of immunity-associated diversifying selection. *J Bacteriol* **193**, 2460–2467.
- Touchon, M. and Rocha, E.P. (2010) The small, slow and specialized CRISPR and anti-CRISPR of *Escherichia* and *Salmonella*. *PLoS One* **5**, e11126. <https://doi.org/10.1371/journal.pone.0011126>.
- Tremblay, C.L., Charlebois, A., Masson, L. and Archambault, M. (2013) Characterization of hospital-associated lineages of ampicillin resistant *Enterococcus faecium* from clinical cases in dogs and humans. *Front Microbiol* **4**, 245.
- Trotta, E. (2014) On the normalization of the minimum free energy of RNAs by sequence length. *PLoS One* **9**, e113380.
- van Schaik, W., Top, J., Riley, D.R., Boekhorst, J., Vrijenhoek, J.E.P., Schapendonk, C.M.E., Hendrickx, A.P.A., Nijman, I.J. *et al.* (2010) Pyrosequencing-based comparative genome analysis of the nosocomial pathogen *Enterococcus faecium* and identification of a large transferable pathogenicity island. *BMC Genom* **11**, 239.
- Wick, R.R., Judd, L.M., Gorrie, C.L. and Holt, K.E. (2017) Unicycler: resolving bacterial genome assemblies from short and long sequencing reads. *PLoS Comput Biol* **13**, e1005595.
- Yosef, I., Goren, M.G. and Qimron, U. (2012) Proteins and DNA elements essential for the CRISPR adaptation process in *Escherichia coli*. *Nucleic Acids Res* **40**, 5569–5576.
- Zhao, X., Yu, Z. and Xu, Z. (2018) Study the features of 57 confirmed CRISPR Loci in 38 strains of *Staphylococcus aureus*. *Front Microbiol* **9**, 1591.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Figure S1. Details of CRISPR types, identical CRISPRs and antibiotic resistance genes of *S. coagulans* isolates. Host D-dog; H-human; Country J-Japan; G-Germany; US-United States of America; SK-South Korea; UK-United Kingdom; AN-Antarctica; IN-India; Strains with identical CRISPR pattern are given the same colour. *MecA*-methicillin resistance gene A (red colour); Presence of antimicrobial resistance gene(s) are indicated with brown/black colour box; *blaZ* and *blaTEM*-Beta lactamase genes; *aac*(6')-aph(2'') and *aadD* – aminoglycoside resistance genes; *lnuA* - lincosamide resistance gene A; *tet*(O) - tetracycline resistance gene O; *erm*(B) – erythromycin resistance gene.

Figure S2. The *cas* genes near the CRISPR loci of *S. coagulans* isolates and their orientations

Figure S3. Genomic pathogenicity islands prediction of *S. coagulans* strain 1031373. Legend: Brown shaded areas inside the circular plot indicate genomic islands. SCCmec, Type IIIA CRISPR–Cas and antimicrobial resistance genes were located in two labelled pathogenicity islands. *blaZ* - Beta lactamase genes; *aac*(6'), *aph*(3'') and *aph*(2'') – aminoglycoside resistance genes; *tet*(O) - tetracycline resistance gene O; *erm*(B) – erythromycin resistance gene. Black trace on the middle circle indicates GC content of the genome. Predicted genomic pathogenicity islands have generally low GC content than the host genome GC content

Supplementary Tables.